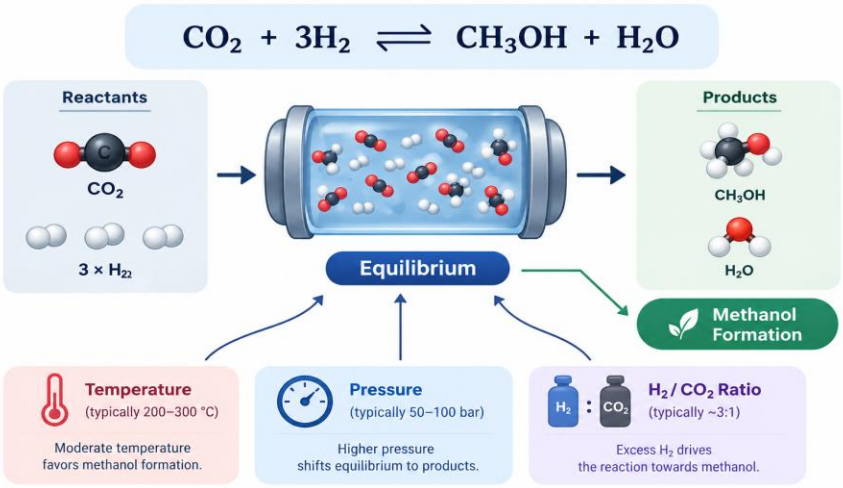
# CHEMICAL PROCESS INTELLIGENCE

## Thermodynamic Modelling, Data Analytics and Process Optimization

### A Computational and Data-Driven Study of $CO_2$ Hydrogenation to Methanol




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**Project Overview:**

1. The project focuses on the **hydrogenation of carbon dioxide to methanol**
2. The reaction involves **$CO_2$ and $H_2$ as reactants** and produces methanol and water
3. At equilibrium, the amount of methanol formed depends on the operating conditions
4. Those variables considered are t**emperature, pressure, and $H_2/CO_2$** feed ratio
5. Thermodynamic principles will be used to understand the behavior and feasibility of the reaction
6. Properties like **enthalpy (ΔH), entropy (ΔS), Gibbs free energy (ΔG), and equilibrium constant (K)** will be calculated
7. An equilibrium model will be developed to evaluate the equilibrium composition and $CO_2$ conversion
8. Python used for calculations, simulate different conditions, and analyze the results
9. SQL and Power BI used to organize, analyze, and visualize the simulation data
10. The overall aim is to **identify thermodynamically favorable operating conditions for $CO_2$ conversion and methanol formation** and demonstrate how chemical thermodynamics can be combined with data analytics for process analysis.

**Problem Statement:**

$CO_2$ hydrogenation to methanol is a well-established equilibrium reaction, but its equilibrium behavior changes significantly with **temperature, pressure, and $H_2/CO_2$ feed ratio**. Although the underlying thermodynamics are known, evaluating and comparing a large number of operating conditions manually is time-consuming.

Therefore, this project develops a **computational thermodynamic framework** to systematically screen different operating conditions and determine their effects on **Gibbs free energy, equilibrium constant, $CO_2$ conversion, and methanol formation**. Python, SQL, and Power BI are used to transform the calculated results into clear operating-condition insights. The study focuses on **thermodynamic screening rather than full industrial process optimization**.

**Research Questions:**

- How does temperature affect the thermodynamic equilibrium of $CO_2$ hydrogenation to methanol?
- How does pressure influence the equilibrium composition and $CO_2$ conversion?
- How does the $H_2/CO_2$ feed ratio affect methanol formation and $CO_2$ conversion?
- How do temperature, pressure, and feed ratio together influence the equilibrium constant and equilibrium extent of reaction?
- Which combinations of operating conditions provide more thermodynamically favorable conditions for methanol formation?
- How can Python, SQL, and Power BI be used to systematically analyze and visualize the calculated operating conditions?

**Objectives:**

The main objective of this project is to **develop a thermodynamic and computational framework to study $CO_2$ hydrogenation to methanol under different operating conditions**.

The specific objectives are:

1. To calculate the enthalpy, entropy, Gibbs free energy, and equilibrium constant of the reaction.
2. To develop an equilibrium model for calculating equilibrium composition and $CO_2$ conversion.
3. To study the effects of temperature, pressure, and $H_2/CO_2$ feed ratio on methanol formation.
4. To use Python to automate thermodynamic calculations and analyze multiple operating conditions.
5. To use SQL to organize, filter, compare, and analyze the generated simulation data.
6. To develop a Power BI dashboard for visualizing thermodynamic and process-performance trends.
7. To identify thermodynamically favorable operating regions for $CO_2$ conversion and methanol formation.

## Chemical System and Reaction:

The $CO_2$ hydrogenation system is represented by two independent reversible reactions.

**Methanol Synthesis (MSR):** with reaction extent $\xi_1$

$$CO_{2\,(g)} + 3H_{2\,(g)} \rightleftharpoons CH_3OH_{(g)} + H_2O_{(g)}$$

**Reverse Water-Gas Shift (RWGS):** with reaction extent $\xi_2$

$$CO_{2\,(g)} + H_{2\,(g)} \rightleftharpoons CO_{(g)} + H_2O_{(g)}$$

**Stoichiometry:** For each species, the number of moles is calculated using:

$$n_i = n_{i0} + \nu_{i1}\,\xi_1 + \nu_{i2}\,\xi_2$$

For the two reactions:

| Species | $n_i$ |
|---|---|
| CO2 | $nCO_{2,0} - \xi 1 - \xi 2$ |
| H2 | $nH_{2,0} - 3\xi 1 - \xi 2$ |
| CH3OH | $nCH_3OH_{,0} + \xi 1$ |
| H2O | $nH_{2}O_{,0} + \xi 1 + \xi 2$ |

| | |
|---|---|
| CO | $nCO,_0 + \xi 2$ |

**Equilibrium:** For each reaction, equilibrium is determined by
$Qp_1 = Kp_1$ and $Qp_2 = Kp_2$
The two extents $\xi_1$ and $\xi_2$ are used to calculate the **equilibrium composition, $CO_2$ conversion, methanol formation, and CO formation**.

## Operating Variables

**Inputs:** Temperature, pressure, and $H_2/CO_2$ feed ratio

**Outputs:** *Kp* equilibrium extents, $CO_2$ conversion, methanol formation, CO formation, and equilibrium composition.

**Thermodynamic Concepts:** The thermodynamic behavior of the reactions is evaluated using enthalpy (ΔH), entropy (ΔS), Gibbs free energy (ΔG), and equilibrium constant ($K_p$). These thermodynamic quantities are calculated for **both MSR and RWGS** at different temperatures and are used to determine their equilibrium behavior.

**Methodology:** A computational thermodynamic approach to analyze $CO_2$ hydrogenation under different operating conditions.

1. Thermodynamic data collection for $CO_2$, $H_2$, $CH_3OH$, $H_2O$, and CO.
2. Calculate ΔH, ΔS, ΔG, and $K_p$ for MSR and RWGS.
3. Vary temperature, pressure, and $H_2/CO_2$ ratio to generate different operating cases.
4. Solve the two equilibrium equations simultaneously to obtain $\xi_1$ and $\xi_2$
5. Calculate equilibrium composition, $CO_2$ conversion, and methanol formation.
6. Use Python for automated calculations and data generation.
7. Use SQL to store, filter, and analyze the generated results.
8. Use Power BI to visualize trends and identify thermodynamically favorable operating regions.

**Thermodynamic Input Data:**

| Chemical Species | Formula (gas) | Standard-ΔH formation | Standard molar entropy |
|---|---|---|---|
| Carbon dioxide | CO2 | −393.51 | 213.79 |

| | | | |
|---|---|---|---|
| Hydrogen | H2 | 0.00 | 130.68 |
| Methanol | CH3OH | | |
| Water vapour | H2O | −241.83 | 188.84 |
| Carbon monoxide | CO | −110.53 | 197.66 |

Source: NIST Chemistry WebBook, Standard Reference Database 69.

**Temperature-Dependent Thermodynamic Properties:**

The thermodynamic properties of gases vary with temperature. Therefore, heat-capacity data are required to calculate the enthalpy and entropy of each species at different temperatures. These properties will be used to determine the temperature-dependent Gibbs free energy and equilibrium constants for both reactions.

The heat capacity at constant pressure is represented as: $C_p^\circ(T) = f(T)$

The enthalpy at temperature (T) is calculated using: $H^\circ(T) = H^\circ(T_{ref}) + \int_{T_{ref}}^{T} C_p^\circ(\mathrm{T})\ \mathrm{dT}$

The standard molar entropy is calculated using: $S^\circ(T) = S^\circ(T_{ref}) + \int_{T_{ref}}^{T} C_p^\circ(\mathrm{T})/(\mathrm{T})\ \mathrm{dT}$

Where (T_ref=298.15K) is the reference temperature. These temperature-dependent properties will be used to calculate the standard Gibbs free energy: $\Delta G^\circ(T) = \Delta H^\circ(T) - T\Delta S^\circ(T)$

The equilibrium constant is then obtained from: $K_p(T) = \exp\left(-\frac{\Delta G^\circ(T)}{RT}\right)$

Data source: NIST Chemistry WebBook, which provides thermodynamic data and heat-capacity information for the chemical species.

**Computational Implementation and SQL Analysis:**

**Python-Based Thermodynamic Model:** Python was used to implement the thermodynamic model and calculate the equilibrium composition of the $CO_2$ hydrogenation system. Cantera provided thermodynamic properties for the chemical species, while NumPy and Pandas supported numerical calculations and data handling. SciPy was used to solve the constrained equilibrium optimization problem, and Matplotlib was used to generate plots.

The model evaluated two independent reactions: methanol synthesis and the reverse water-gas shift reaction. The equilibrium reaction extents were determined by minimizing the deviations between the calculated reaction quotients and their corresponding equilibrium constants, subject to non-negative species mole constraints.

**Operating Condition Analysis:** The model was evaluated at a baseline condition of 500 K and 50 bar, using an initial feed of 1 mol $CO_2$ and 3 mol $H_2$. The analysis was then extended to five temperatures (400, 450, 500, 550, and 600 K) and four pressures (10, 30, 50, and 70 bar), giving 20 operating conditions.

For each condition, the model calculated equilibrium species amounts, $CO_2$ conversion, methanol yield, and carbon monoxide formation. The results were exported to CSV files for further analysis.

**MySQL-Based Data Analysis:** A MySQL database named `methanol_project` was created to store the equilibrium results in a structured table called `equilibrium_results`. The dataset included temperature, pressure, $CO_2$ conversion, methanol yield, methanol amount, carbon monoxide amount, and solver status.

SQL queries were used to rank operating conditions by methanol yield and summarize model results. This enabled systematic comparison of the tested operating conditions and prepared the results for visualization.

## Results and Discussion

**Baseline Equilibrium Results:** The baseline equilibrium calculation was performed at 500 K and 50 bar using an initial feed of 1 mol $CO_2$ and 3 mol $H_2$. The temperature-dependent thermodynamic model provided the equilibrium constants for both methanol synthesis and the reverse water-gas shift reaction.

At these conditions, the calculated equilibrium extents were approximately 0.2391 mol for methanol synthesis and 0.04183 mol for the reverse water-gas shift reaction. The resulting methanol amount was approximately 0.2391 mol.

The calculated $CO_2$ conversion was 28.10%, while the methanol yield was 23.91%. These results indicate that a portion of the initial $CO_2$ was converted into methanol, while the competing reverse water-gas shift reaction produced carbon monoxide.

**Effect of Temperature on Methanol Yield:** The model results showed a decrease in methanol yield as temperature increased across the tested range. At a given pressure, the highest methanol yields generally occurred at the lower temperatures examined.

This trend is consistent with the exothermic nature of methanol synthesis. According to Le Chatelier's principle, increasing temperature shifts the equilibrium of an exothermic reaction toward the reactants, reducing the equilibrium amount of methanol.

**Effect of Pressure on Methanol Yield:** At a fixed temperature, increasing pressure generally increased the equilibrium methanol yield in the tested conditions. Methanol synthesis consumes four moles of gaseous reactants and produces two moles of gaseous products. Consequently, higher pressure favors the product side of this reaction under the ideal-gas equilibrium model.

The combined effects of temperature and pressure demonstrate that operating conditions strongly influence the predicted equilibrium composition.

**Comparison of Operating Conditions:** Among the 20 temperature–pressure combinations examined, the highest calculated methanol yield was approximately 62.83%, obtained at 400 K and 70 bar. The maximum calculated $CO_2$ conversion was approximately 62.87% at the same condition.

These values identify the best-performing condition within the tested range according to the equilibrium model. They should not be interpreted as a validated industrial optimum because catalyst kinetics, reactor design, heat and mass transfer, separation, and process economics were not included in the analysis.

**Power BI Dashboard and Visualization:**

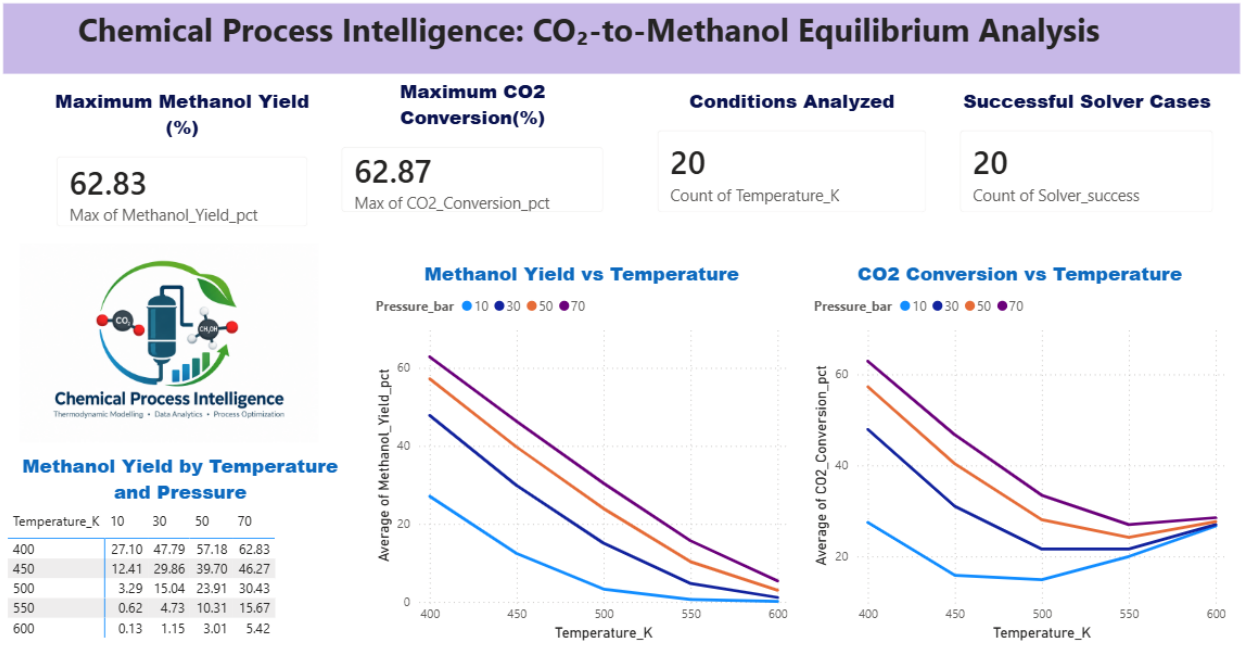



Figure 1. Power BI dashboard for equilibrium analysis of $CO_2$ hydrogenation to methanol.

Microsoft Power BI was used to visualize the equilibrium results obtained from the Python model. The calculated dataset containing 20 temperature–pressure combinations was imported into Power BI to develop a consolidated performance dashboard.

The dashboard includes four key performance indicators: maximum methanol yield, maximum $CO_2$ conversion, total operating conditions analyzed, and successful equilibrium solver cases. These indicators provide a concise overview of the model results.

Two line charts illustrate the effects of temperature and pressure on methanol yield and $CO_2$ conversion. A temperature–pressure matrix presents the calculated methanol yield for each operating condition, allowing direct comparison of the results.

The dashboard integrates thermodynamic modelling with data analytics and provides an accessible way to interpret the computational results. It supports the identification of favorable equilibrium operating regions within the tested conditions.

**Model Validation and Limitations:** The equilibrium solver was evaluated across 20 temperature–pressure combinations. All 20 cases reported successful solver status. The calculated equilibrium compositions were checked against the non-negativity constraints, and the reaction quotients were compared with their corresponding equilibrium constants.

The numerical results were also organized in a MySQL database and reviewed through SQL queries and Power BI visualizations. This provided consistency checks across the computational, database, and visualization stages.

Successful numerical convergence indicates that the solver found solutions satisfying the implemented equilibrium conditions. However, it does not independently establish experimental accuracy or validate the model against industrial operating data.

he present study has several limitations:

1. **Ideal-gas assumption:** Gas-phase species were treated as ideal gases. At elevated pressures, real-gas non-ideality may influence equilibrium predictions.
2. **Equilibrium-based analysis:** The model predicts thermodynamic equilibrium compositions and does not calculate reaction rates or the time required to reach equilibrium.
3. **Catalyst effects:** Catalyst activity, selectivity, deactivation, and kinetic parameters were not included.
4. **Reactor and process effects:** Heat transfer, mass transfer, reactor design, separation, recycle, and process economics were outside the scope of the model.

5. **Validation data:** The results were checked for numerical consistency, but independent experimental or industrial data were not used to validate the predictions.

Therefore, the results should be interpreted as equilibrium-based estimates for the specified feed and operating conditions rather than direct predictions of industrial reactor performance.

**Conclusion:** This project developed a computational and data-driven framework for studying $CO_2$ hydrogenation to methanol through thermodynamic modelling, SQL-based data analysis, and Power BI visualization.

A Python model was implemented to evaluate two independent reactions: methanol synthesis and the reverse water-gas shift reaction. Temperature-dependent thermodynamic properties were used to calculate equilibrium constants and estimate equilibrium compositions under different operating conditions.

At the baseline condition of 500 K and 50 bar, the model predicted a methanol yield of 23.91% and a $CO_2$ conversion of 28.10%. Across the 20 temperature–pressure combinations investigated, the highest calculated methanol yield was approximately 62.83%, with a maximum $CO_2$ conversion of approximately 62.87%, at 400 K and 70 bar.

The calculated results were stored in MySQL and analyzed using SQL queries. Power BI visualizations enabled comparisons of methanol yield and $CO_2$ conversion across the tested conditions. The integration of thermodynamic modelling, database analysis, and visualization demonstrated how computational tools can support the interpretation of chemical process behaviour.

Overall, the study identified favorable equilibrium operating regions within the specified range and established a foundation for further process analysis. The findings remain subject to the assumptions and limitations of the equilibrium model.

**Future Scope:** he present study can be extended in several directions:

1. **Real-gas modelling:** Incorporate fugacity coefficients and suitable equations of state to improve equilibrium predictions at elevated pressures.
2. **Reaction kinetics:** Integrate catalyst-specific kinetic models to estimate reaction rates and reactor performance.
3. **Reactor simulation:** Extend the equilibrium model to plug-flow or other reactor models that account for heat and mass transfer.

4. **Process optimization:** Investigate a broader operating range while considering energy requirements, selectivity, separation, recycle, and economic constraints.
5. **Experimental validation:** Compare predicted equilibrium compositions with published experimental or industrial data.
6. **Advanced analytics:** Extend the database and dashboard to support additional process variables, sensitivity analysis, and automated reporting.



Thank You